

**Lab Goal :** This lab was designed to teach you more about graphs, mazes, and recursion.

**Lab Description :** A maze is a two dimensional structure. All of the mazes will have the entrance point at the top left corner. The exit point will be the bottom right corner.

You will find the minimum number of bombs that must be used go from the entrance to the exit. You have an unlimited number of bombs, each of which can blow up one wall. You want to use the least amount of bombs possible. You may not need to use any bombs.

**Helpful Hints / Assumptions:** All input characters will be '#'s and '.'s, representing walls and open spaces respectively. You can travel through open spaces '.'s but not walls '#'s.

### Sample Data :

```
5
..###
.#..#
#####
#.#..
###..
9
..#####
...#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#####
9
..#####
...#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#####
```

#### Files Needed ::

You must create the java file.

The data file is provided.  
maze\_bombs.dat

#### algorithm help

```
if ( r and c are in bounds and current spot is a 1 )
    mark spot as visited
if we are at the exit
    mark exit found as true
else
    4 recursive calls up down left right
```

### Sample Output :

```
..###
.#..#
#####
#.#..
###..
2 bomb(s)
..#####
...#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#####
3 bomb(s)
..#####
...#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#.#.#.#
#####
1 bomb(s)
```